Department Of הנדסת תוכנה

Course No.	3502811			מספר קורס
Course Name	INTRODUCTION TO ALGORITHMS	מבוא לאלגוריתמים		שם הקורס
Lecturers' Name	Zrihen Shaked, Levy Avivit	שקד זריהן, אביבית לוי	גב', פרופ'	מרצי הקורס
Year	2026	תשפו		שנת הוראה
Credits	4			נקודות זכות
Prerequisite	(Data Structures (3502810	מבני נתונים (3502810)		דרישות קדם

Course Summary

The course objective is to teach the foundations of the design of algorithms. The course gives the students tools and techniques in basic and advanced algorithms with an emphasis on knowledge of a variety of subjects and fields and on analysis and .performance evaluation

Main Teaching Methods

Lecture / Studio / Lab	Frontal
Tutorial / Practice	Frontal

Learning Outcomes

Learning Outcomes

- understand the importance of the design of algorithms in programming and Software development
- be familiar with the basic notions in algorithms
- be familiar with basic algorithm analysis
- be familiar with performance valuation techniques
- apply this knowledge to various problems

Course Outcomes

Assessment Structure

- 80% - מבחן.
- 10% - מטלה. Notes: הגשת תרגילי בית מדי שבוע
- 10% - מבחן. Notes: בוחן אמצע על תרגילי הבית,

בוחן האמצע הוא חובה ואין מועד ב.

Student Obligations	According to Shenkar's Regulations.

Course Structure

?Asynchronous	Notes	Topic	Week/Meeting/ Date
		Introduction: polynomial vs. intractable algorithms and problems	1
	Assignment 1	Graphs algorithms: Strongly connected components	1
	Assignment 2	Graphs algorithms: Minimum spanning tree and (Greedy Algorithms - Prim	2
	Assignment 3	Minimum spanning tree - Union-Find forest data structure and Kruskal algorithm	3
	Assignment 4	Graphs algorithms: Single source shortest paths - Dijkstra	4
	Assignment 5	Graphs algorithms: Single source shortest paths - Belman-Ford and difference constraintsd	5
	Assignment 6	All-pairs shortest paths: Dynamic programming and Floyd-Warshall algorithm	6
	Assignment 7	Flow networks: Ford-Fulkerson method (If possible: Edmonds-Karp algorithm), Max-flow-Min-cut Theorem. (If possible: Maximum matching in (bipartite graphs	7
	Assignment 8	String matching algorithms: naïve, Rabin-Karp	8
	Assignment 9	String matching algorithms: KMP, string matching using convolutions	9
	Assignment 10	Polynomials multiplication and FFT	10
		Polynomials multiplication and FFT	11
	Assignment 11	Introduction to randomized algorithms: Monte-Carlo and Las-Vegas algorithms,	12

	an2alysis of random choice of pivots in Quicksort algorithm		
13	Introduction to randomized algorithms: Karger algorithm for finding min-cut in graphs (If possible: Skip-list data (structure		

Bibliography / Filmography

מספר	הפריט הביבליוגרפי	חובה / רשות
	Introduction to Algorithms. Cormen T. H., Leiserson C. E., Rivest R. L., Stein C., MIT, 2022	
	Randomized Algorithms, R. Motwani and P. Raghavan, Cambridge University Press, 2010	